Palak Deb Patra

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EDUCATION

National Institute of Technology, Rourkela — **Dec 2021 – Jun 2025**
Bachelor of Technology in **Computer Science & Engineering** | **CGPA: 8.52/10** — **Rourkela, India**

PROFESSIONAL EXPERIENCE

NatWest Group | **Software Engineer** — **Jul 2025 – Present**
Python, FastMCP, Azure, Microsoft Copilot Studio, Power Platform — **Bangalore, India**

Project: Colleague Learning Assistant (CLA)

- Built **end-to-end AI observability** using **OpenTelemetry** and **Azure Application Insights** to trace requests, tool calls and conversations, with **8 failure categories.**
- Developed **131 automated tests**, **11 Kusto queries** for agent validation and monitoring activity.
- Configured **single sign-on (SSO)** using **PingFederate** and **Microsoft Entra ID**, with a **three-step fallback** to identify employees and handle missing authentication details.
- Integrated **12 months of Workday learning completion records** to personalise course recommendations
- Built an **Azure Blob Storage pipeline** for GitLab metrics with change detection and fallback to the last valid copy.
- Redesigned **3 Microsoft Teams widgets** for Learning Courses, ECDF progress and GitLab metrics to follow NatWest branding guidelines and resolve rendering issues.

Project: ChLoE

- Worked on migrating deployments from **Power Platform pipelines** to **GitLab CI/CD** in line with bank security requirements.
- Improved Power Platform FAQ knowledge sources and evaluated revised agent responses in SIT and UAT environments, with **23 of 26 test cases passing**.
- Fixed **duplicate agent responses** after deployments and corrected the Microsoft Teams **"Start over" trigger** to clear chat history, close active topics and restart conversations.

NatWest Group | **Software Engineering Intern** — **May 2024 – Jul 2024**
Python, GraphQL, REST APIs, Snowflake — **Gurgaon, India**

Project: OSPO Metrics

- Collected and consolidated **engineering activity metrics** for a business unit using **GitLab GraphQL and REST APIs**, and automated data extraction and CSV exports in **Python**, reducing manual processing time by **28%**.
- Loaded **16 datasets** into **Snowflake** and developed a unified data model for interactive **ThoughtSpot dashboards**.

PERSONAL PROJECTS

CodeShift | Python, LangGraph, Ollama, TypeScript | GitHub — **Aug 2026**

- Built a **six-agent system** using LangGraph and Ollama to migrate Python projects to TypeScript, converting supporting modules before the files that use them.
- Used **mypy** and **tsc** for type checking and developed **differential tests** to compare outputs, exceptions and object states between the original and translated code.
- Evaluated the system on a **31-module accounting application**, testing **197 functions and methods** and confirming matching behaviour for **15 modules** under the generated tests.

Memex | Python, LangGraph, ChromaDB, Ollama, Streamlit | GitHub — **Aug 2026**

- Built a **local RAG system** using **dense retrieval, BM25 and cross-encoder reranking**, with support for source citations, OCR and automatic index updates, and indexed **3,050 files** into **42,743 chunks** in approximately **15 minutes** on a local GPU.
- Evaluated retrieval on **42 questions** across a separate **23-document corpus**, ranking a relevant result first for **41 of 42 questions** and achieving **97.6% Hit@1**, **0.988 MRR** and **100% Recall@5**.

TECHNICAL SKILLS

Languages	Python, Java, C++, TypeScript, SQL
AI & ML	PyTorch, TensorFlow, scikit-learn, OpenCV, spaCy, NLTK, NVIDIA CUDA
LLMs & Agents	LangGraph, LangChain, Hugging Face, Ollama, FastMCP, Claude, Codex
Web & Databases	Angular, Streamlit, FastAPI, REST APIs, GraphQL, Snowflake, Oracle, MySQL, ChromaDB
Cloud & Tools	AWS, Azure, Git, GitLab CI/CD, Ansible, Jupyter

ACHIEVEMENTS

- Selected for the **Amazon ML Summer School 2023**.
- Solved **800+** LeetCode problems with a peak rating of **1795**, ranking in the **top 8%** globally.